Thermodynamic description of the Cr–Fe system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cr–Fe binary system as calculated with pycalphad from the SGTE dataset CrFe-11Xio.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, σ → (Cr,Fe)1 + (Cr,Fe)2, lies at 790 K.

1. Introduction

The Cr–Fe system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic, gamma loop, solid miscibility gap. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of W. Xiong, P. Hedstrom, M. Selleby, J. Odqvist, M. Thuvander, Q. Chen, Calphad, 35, 355-66(2011). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Cr melts at 2180 K. Pure Fe melts at 1818 K.

2. Phase diagram and invariant reactions

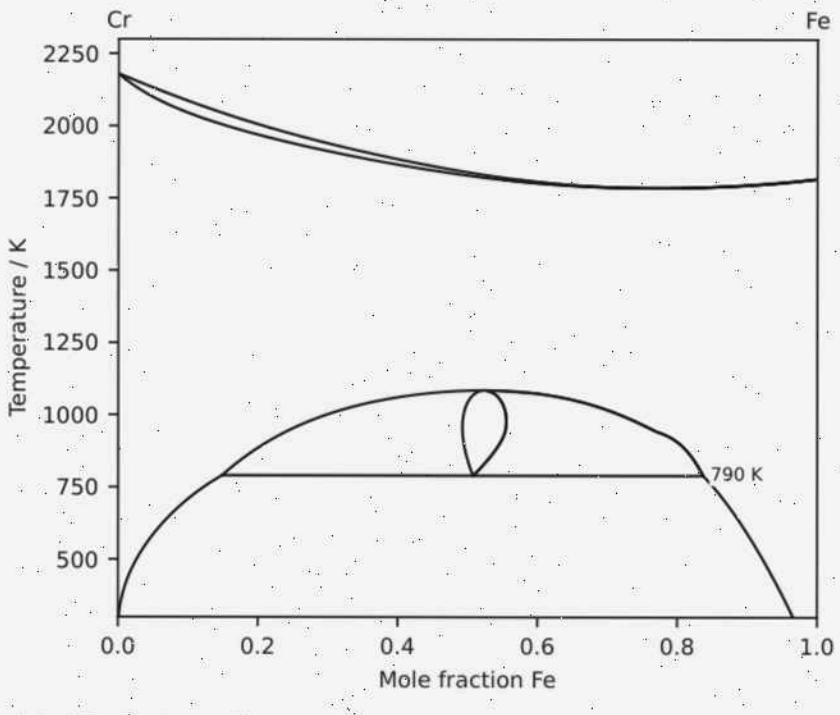


Fig. 1. Calculated Cr–Fe phase diagram.

Eutectoid equilibrium σ → (Cr,Fe)1 + (Cr,Fe)2 is found at 790 K and 50.8 at.% (σ).

Table 1. Invariant reactions and congruent points, calculated (Cr–Fe).

Reaction	Type	T / °C	Compositions (x(Fe))		
		this work	phase 1	phase 2	phase 3
(Cr,Fe) → σ	congruent transformation	812.5	0.522	0.522	–
σ → (Cr,Fe)1 + (Cr,Fe)2	eutectoid	517.0	0.508	0.149	0.837

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Cr–Fe).

Phase	Pearson	SB	Prototype	T min (K)	T max (K)	Range (x(Fe))	T (K)
(Cr,Fe)	cI2	A2	W	< 300	2180.0	0.000–1.000	1640
σ	tP30	D8b	σ-CrFe	790.2	1085.6	0.493–0.555	970

Structure labels follow the TDB phase names (– where the name does not identify a structure). Stability range: temperatures between which the phase is stable in some part of the diagram.

According to Table 2, σ remains stable up to 1086 K (congruent transformation).

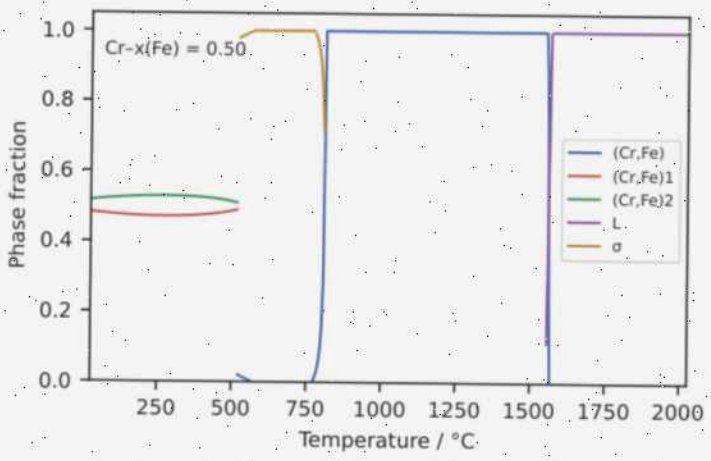


Fig. 2. Calculated equilibrium phase fractions versus temperature for Cr–x(Fe) = 0.50.

4. Thermodynamic properties

At 2250 K the integral mixing enthalpy of the liquid has its maximum, 37 J/mol, at 78.0 at.% Fe. The calculated enthalpy of formation of σ at 298.15 K is 5.89 kJ/mol per mole of atoms. Its heat capacity at room temperature is 24.12 J/(mol K). For the equiatomic liquid at 2250 K, the Gibbs energy of mixing is −16.33 kJ/mol.

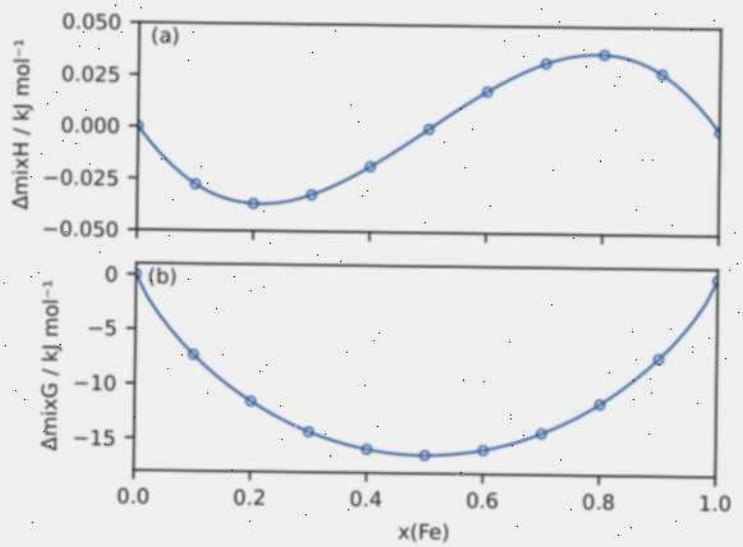


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2250 K.

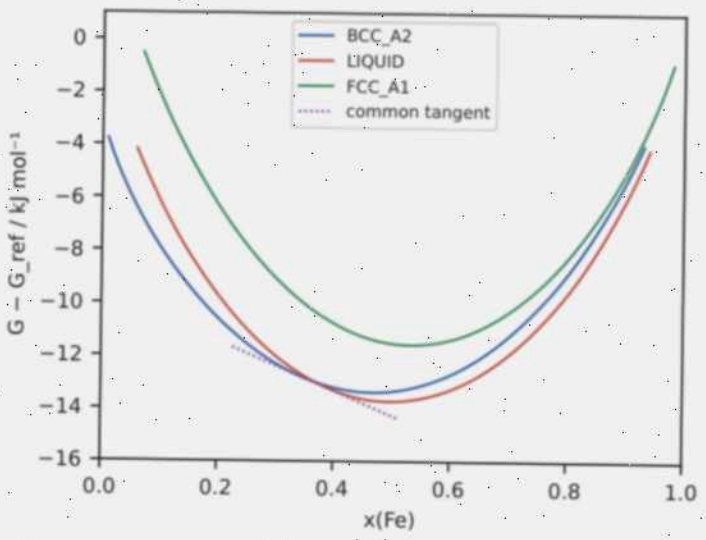


Fig. 4. Calculated Gibbs energy curves at 1890 K.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Cr–Fe).

		298.15 K, per mole of atoms		
Phase	**at.% Fe**	**ΔfH / J mol⁻¹**	**ΔfS / J mol⁻¹ K⁻¹**	**Cp / J mol⁻¹ K⁻¹**
σ[a]	52.0	5889 ± 1000	9.10	24.12

Reference: stable elements (SER) at 298.15 K; values per mole of atoms. ± values are synthetic illustration uncertainties, not data.

[a] Not stable at 298.15 K in this description; properties of the metastable phase.

Table 4. Integral and partial mixing properties of the liquid at 2250 K (Cr–Fe).

at.% Fe	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Cr)	a(Fe)	ΔH(Cr) / J mol⁻¹	ΔH(Fe) / J mol⁻¹
10	-28	-7321	0.8931	0.0553	-10	-187
20	-37	-11552	0.7759	0.1259	-34	-49
30	-32	-14287	0.6539	0.2113	-62	38
40	-18	-15840	0.5323	0.3101	-86	83
50	0	-16333	0.4155	0.4198	-96	96
60	18	-15803	0.3073	0.5372	-83	86
70	32	-14222	0.2104	0.6583	-38	62
80	37	-11478	0.1265	0.7787	49	34
90	28	-7265	0.0564	0.8940	187	10

Reference states: pure liquid Cr and Fe at 2250 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] W. Xiong, P. Hedstrom, M. Selleby, J. Odqvist, M. Thuvander, Q. Chen, Calphad, 35, 355-66(2011).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.